



PERSONAL DETAILS

Name : MOHD MURTADHA BIN MOHAMAD
Gender : MALE
Date of birth : 16 AUGUST 1977
Nationality : MALAYSIAN
Marital Status : MARRIED
Permanent Address : 66 JALAN MURNI TAMAN SURIA 81100 JOHOR BAHRU

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UNIVERSITI TEKNOLOGI MALAYSIA,
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Expertise : Robot motion planning, acoustic sensor network, indoor positioning

ACADEMIC QUALIFICATIONS

Year 2006 : Ph.D. Electrical Engineering
Heriot-Watt University, United Kingdom
Year 2002 : M.Sc. Embedded System
Heriot-Watt University, United Kingdom
Year 1999 : B. Eng. Computer
Universiti Teknologi Malaysia

AWARD AND HONORS RECEIVED

1. 2025 Award of Excellent Service, Faculty of Computing Universiti Teknologi Malaysia for year 2024

2. 2025 Citra Karisma Award of Author in Indexed Journal in the Category Centre of Excellence (COE) Institute for Smart Infrastructure and Innovative Construction (ISIIC) for year 2024
3. 2025 Award of Appreciation for WoS Q1/Q2 Journal Publication, Faculty of Computing, Universiti Teknologi Malaysia, as First UTM Corresponding Author
4. 2024 Bronze Award, 26th Industrial Art and Technology Exhibition & Competition (INATEX 2024)
5. 2024 Award of Excellent Service, Faculty of Computing Universiti Teknologi Malaysia for year 2023
6. 2022 Award of The Best Consultation (Individual), Faculty of Engineering, Universiti Teknologi Malaysia 2021
7. 2021 Award of Excellent Service, Universiti Teknologi Malaysia 2020
8. 2018 Award of Excellent Service, Centre of Information Communication Technology 2017
9. 2017 Award of Excellence, Active Blended Learning Course, Semester 2-2016/2017
10. 2014 Staff Honors Award for achieving over 90% of Annual Employee Performance Evaluation (e-LPPT)
11. 2014 Winner of *Pertandingan Memasyarakatkan Inovasi 2014 Zon Selatan*, MySEL-242 Agrosquare with the awarded grant of RM50,000.
12. 2012 Erasmus Mundus IDEAS 2012 Scholarship for Staff Mobility at Laboratory of Software Quality, Universiteit Technische Eindhoven, Netherlands with awarded grant of EUR10,000
13. 2008 UTM Excellent Service Award 2007
14. 2006 Salary Increment Prize for Employee in Service Schemes for Various Qualification Entry Under Education Service Classification

PROFESSIONAL MEMBERSHIP / QUALIFICATIONS / RECOGNITION

1. Member of Institute of Electrical Electronic Engineers (IEEE) Membership No.: 41536103, 2002 – recent.
2. Member of World Academy of Science , Engineering and Technology, Scientific and Technical Committee & Editorial Review Board on Electronics and Communication Engineering Member of World Academy of Science, Engineering and Technology (WASET), 2005
.
3. Member of International Association of Engineers (IAENG), Membership No.: 136990, 2014 – recent.
4. Member of Society of Digital Information and Wireless Communication (SDIWC), Membership no.: 1853, July 2014 – recent.
5. Sijil Talaqqi Bersanad Riwayat Hafz 'An 'Asim 30 Juzu' AlQuran secara Tariq Al-Syatibi.
6. Siji Talaqqi Bersanad Riwayat Syu'bah 'An 'Asim 30 Juzu' AlQuran secara Tariq Al-Syatibi
7. Entry Level Scientific Diver, International Technical Diving Agency ITDA Europe, Membership no.: 196373, July 2009 – July 2011.

ADMINISTRATIVE EXPERIENCE

Faculty Level

1. Current – Coordinator, Doctor of Philosophy (Computer Science) Programme, Faculty of Computing, UTM.
2. Former Research Manager, Faculty of Computing, UTM.
3. July 2022 – Acting Research Manager, School of Computing, Faculty of Engineering, UTM.
4. Former Head, Pervasive Computing Research Group, Faculty of Computing, UTM.
5. 2011 - Committee of First Year Experience of Faculty of Computer Science and Information System
6. 2011 - Committee of Internationalisation of Faculty of Computer Science and Information System

7. 2010 - Committee of First Year Experience of Faculty of Computer Science and Information System
8. 2010 - Committee of Internationalisation of Faculty of Computer Science and Information System
9. Jan. 2007 - Dec. 2008 Committee of Faculty of Computer Science and Information System Student Development Program

University Level

1. Former Deputy Director (Application Development Management), UTMDigital, UTM.
2. Sept. 2015 – March 2018 – Academic Fellow of Centre of Information Communication Technology, UTM
3. 1 November 2012 – 31 December 2015 – Residential College Principal of Kolej Perdana
4. 1 September 2007 – 31 August 2012 – Residential College Principal of Kolej Tun Ghafar Baba
5. April 2014 - Auditor of Occupational Safety and Health Award 2013
6. 30 Oct. - 7 Nov. 2008 - Internal Auditor of Biomedical and Health Science Faculty

OTHERS EXPERIENCE

UNIVERSITY APPOINTMENT/COMMITTEE

1. Appointment as Research Fellow, Centre for The Study of Built Environment In The Malays World (KALAM), Universiti Teknologi Malaysia – 1st January 2024 – 31st December 2026.
2. Appointment as Deputy Chair UTM E-Learning Committee, UTMLead – 1st October 2018 – 30 September 2020.
3. Appointment as Committee Member UTM Open Distance Learning (ODL), 9 September 2018 - September 2020.
4. Appointment as Member of Standing Committee of the Senate of Academic Curriculum and Quality – 1 April 2018 – 31 March 2021.

5. Appointment as Technical Evaluation Committee Members for ICT Acquisition Universiti Teknologi Malaysia – 1 July 2018 – 30 June 2020
6. Appointment as Information System Technical Committee Members for SINERGI UTM 4.0 Universiti Teknologi Malaysia – 8 May 2018 – 30 September 2018
7. Appointment as Technical Evaluation Committee Members of Information Technology Award Category In Citra Karisma UTM 2018 – March 2018
8. Appointment as Technical Evaluation Committee Members for CICT WIFI Tender Project – 15 April 2018 – 14 April 2019.
9. Appointment as Member of Human Resource Development Panel, Centre of Information Communication Technology (CICT) UTM – 1 April 2018- 31 December 2019.
10. Appointment as Committee Members for UTM Library Quick Response (QR) Code Project – May 2018.
11. Appointment as Technical Evaluation Committee Members Tender PTJ UTM.B 24/2015 Tender to supply Testing And Commissioning Of General Applications Servers And Storages For High Performance Computing System In Universiti Teknologi Malaysia - 4 September 2015 – 3 September 2016
12. Appointment as Member of the Standing Committee of the Senate of Student Development and Alumni May 2014 – Sept. 2016
13. Appointment as Main Committee of Communication of Behavioral Impact (COMBI) of Resident College, Universiti Teknologi Malaysia – 11 May 2015 -10 May 2016
14. Appointment as Student Discipline Committee Panel – 19 November 2015.

NATIONAL APPOINTMENT/COMMITTEE

1. Appointment as Director Persidangan Pembangunan Pelajar Kebangsaan NASDEC 2018, Malaysian Ministry of Education (MOE), Malaysian Department of Higher Education (JPT) September 20 – 22 2018
2. Appointment as Single Point of Contact (SPoC) for Infosys Campus Connect Programme in collaboration with Malaysian Development Corporation (MDec), Malaysian Department of Higher Education (JPT) and Infosys Limited, India, July 2012 - July 2014

INTERNATIONAL APPOINTMENT/COMMITTEE

1. Technical Review Committee of 5th International Conference on Computer Applications (ICCAA 2015), 24 – 26 December 2015, Pondicherry, India
2. Technical Programme Committee of 4th International Visual Informatics Conference (IVIC'15), 17 – 19 November 2015, Putrajaya, Malaysia
3. Organizing Programme Committee of International Conference on Computational and Social Sciences (ICCSS-2015), 25 – 27 August 2015, Johor Bahru, Malaysia
4. Technical Committee Member The Third International Conference on Human Machine Interaction 2015
5. Appointment as Technical Review Committee Member of International Conference on eBusiness, eCommerce, eManagement, eLearning, and eGovernance (IC5E2015), 24 – 25 July 2015, London, UK
6. Program Committee for 13th International Conference on Intelligent Software Methodologies, Tools, and Techniques (SOMET14), 22 – 24 September 2014, Langkawi, Malaysia
7. Technical Program Committee IEEE International Symposium on Biometrics & Security Technologies (ISBAST2014), 26 – 27 August 2014, Kuala Lumpur, Malaysia
8. Program Committee 5th Asian Conference on Intelligent Information and Database Systems (ACIIDS2013), 18 – 20 March 2013, Kuala Lumpur, Malaysia
9. Technical Program Committee 3rd International Visual Informatics Conference 2013 (IVIC2013), September 2013, Selangor, Malaysia
10. Technical Program Committee 2013 International Conference on IEEE RFID Technical and Application, September 2013, Johor Bahru, Malaysia

INTERNATIONAL JOURNAL REVIEWER

1. Reviewer of "Securing Smart Homes through SD-WAN: A Software-Defined Network Approach" for the IEEE Access, 2025
2. Reviewer of "Electromagnetic Environment Simulation Software Design and Its Application Using DRFM Techniques on Software Defined Radio" for the IEEE Access, 2025

3. Reviewer of "Performance Improvement of Energy Criterion Method Based on the Instantaneous Phase Characteristics in Wi-Fi Transient Detection for RF Fingerprinting" for the IEEE Access, 2025
4. Reviewer of "AAPDEM: An Intelligent Model for Automated Attack Path Discovery and Exploitation in Modern Networks" for the IEEE Access, 2025
5. Reviewer of "SITRUS: Simulation and Analysis of Irregular Tessellated Wireless Sensor Networks Using Cooja" for the IEEE Access, 2025
6. Reviewer of "Towards Intelligent Caching in DSR Protocols: Route Compression, Hierarchical Structures, and Link-Aware Optimization in MANETs" for the IEEE Access, 2025
7. Reviewer of "A Survey of Challenges and Opportunities in Vertical Federated UnLearning" for the IEEE Access, 2025
8. Reviewer of "Meta-Reinforcement Learning for Multi-Energy Building Microgrids: Curriculum Design, Rapid Adaptation, and Scalable Deployment" for the IEEE Access, 2025
9. Reviewer of "Dynamic Distributionally Robust Optimization with Adaptive Ambiguity Sets: A Machine Learning-Enhanced Framework for Market Decision-Making" for the IEEE Access, 2025
10. Reviewer of "Trust-Aware and Energy-Efficient Clustering and Routing Using Modified BBO for 6G-IoT Networks" for the IEEE Access, 2025
11. Reviewer of "TriageHD: A Hyper-Dimensional Learning-to-Rank Framework for Dynamic Micro-Segmentation in Zero-Trust Network Security" for the IEEE Access, 2025
12. Reviewer of "Multi-Agent Deep Reinforcement Learning for Dynamic Routing in MANETs Using Graph Neural Networks" for the IEEE Access, 2025
13. Reviewer of "Energy-Aware MARL for Coordinated Underwater Sensing and Data Collection" for the IEEE Access, 2025
14. Reviewer of "GAT-APG: Graph Attention Network-Based Attack Path Generation for Security Simulation" for the IEEE Access, 2025
15. Reviewer of "Federated Learning-Based Trust Evaluation with Fuzzy Logic for Privacy and Robustness in Fog Computing" for the IEEE Access, 2025
16. Reviewer of "Adaptive Federated Learning for Robust PV Power Forecasting Under Weather-Induced Concept Drift" for the IEEE Access, 2025
17. Reviewer of "A Study on Robust Control Scheme Using Prescribed Performance-based Time Delay Control and RBF Neural Network" for the IEEE Access, 2025

18. Reviewer of "Interaction Framework and Method of Electric Vehicles Aggregator and Distribution Transformer Area Based on Federated Learning" for the IEEE Access, 2025
19. Reviewer of "Mind the Gap: A Survey of Challenges and Opportunities in Vertical Federated UnLearning" for the IEEE Access, 2025
20. Reviewer of "Research on Deep Learning-Based Optimization Algorithms for IoT Communication" for the IEEE Access, 2025
21. Reviewer of "Spreading Factor Allocation in LoRaWAN Through Multi-objective Optimization" for the IEEE Access, 2025
22. Reviewer of "Leveraging Federated and Incremental Learning in Spatial Crowdsourcing: A Comprehensive Review" for the IEEE Access, 2025
23. Reviewer of "New Methodology to Calibrate Depth Sensors in Multi-Modal Dynamic Setups" for the IEEE Access, 2025
24. Reviewer of "Privacy Preservation in Decentralized Asynchronous Federated Learning for Substation Scenarios" for the IEEE Access, 2025
25. Reviewer of "Dynamic State Consistency in Software Defined Networks: A Multi-Controller Approach" for the IEEE Access, 2025
26. Reviewer of "FME-GCN: A Multi-Source Heterogeneous Data Fusion Approach for High-Accuracy Traffic Flow Prediction" for the IEEE Access, 2025
27. Reviewer of "AnomLocal: A Hybrid Local-Global Anomaly Detection Model for Network Security Using Federated Learning" for the IEEE Access, 2025
28. Reviewer of "Enhancing Anomaly Detection in Physical Activity Motivation: An Educational Perspective" for the IEEE Access, 2025
29. Reviewer of "Reinforcement Learning-Assisted Secure Reliable Underwater Wireless Acoustic Communications" for the IEEE Access, 2025
30. Reviewer of "Crew Scheduling Problems: A Systematic Literature Review" for the IEEE Access, 2025
31. Reviewer of "Secure Data Transmission Against Cooperative Blackhole Attack in Mobile Ad-Hoc Network (MANET)" for the IEEE Access, 2025
32. Reviewer of "Federated Learning for Intelligent and Sustainable Environmental Monitoring: A Comprehensive Review" for the IEEE Access, 2025
33. Reviewer of "Decoding MANET: A Systematic Review" for the IEEE Access, 2025

34. Reviewer of "Machine Learning in e-Commerce: Trends, Applications, and Future Challenges" for the IEEE Access, 2025
35. Reviewer of "A Hybrid CT-DEWCA Based Energy-Efficient Routing Protocol for Data and Storage Nodes in Underwater Acoustic Sensor Networks" for the IEEE Access, 2025
36. Reviewer of "Integrating Machine Learning and Multi-Objective Optimization in Biofuel Systems: A Review" for the IEEE Access, 2025
37. Reviewer of "IP SafeGuard - An AI-Driven Malicious IP Detection Framework" for the IEEE Access, 2025
38. Reviewer of "Dynamic Pricing Models in E-Commerce: Exploring Machine Learning Techniques to Balance Profitability and Customer Satisfaction" for the IEEE Access, 2025
39. Reviewer of "Collaborative Intrusion Detection Using Federated Deep Learning and Quantum Smart Contracts on Ethereum Blockchain for Enhanced Cloud Security" for the IEEE Access, 2025
40. Reviewer of "A Dynamic Trust Evaluation and Prediction Model for WSN Based on Multiple Indexes" for the IEEE Access, 2025
41. Reviewer of "A Framework for Enhanced File Sharing Security: Integrating Neural Networks, Deep Reinforcement Learning and Knot Theory" for the IEEE Access, 2025
42. Reviewer of "MULTI-ATTACK DETECTION USING FORENSICS AND ADAPTIVE DUAL-CHANNEL PULSE COUPLED NEURAL NETWORK BASED PREVENTION FOR SECURE IOT-MANETS" for the Australian Journal of Forensic Sciences, 2025
43. Reviewer of "ADVANCED MULTI-AGENT T03 DEEP REINFORCEMENT LEARNING FOR ROBUST AUV NAVIGATION AND DEPTH CONTROL" for the Automatika: Journal for Control, Measurement, Electronics, Computing and Communications, 2025
44. Reviewer of "REAL-TIME DETECTION OF BLACK HOLE ATTACKS IN MANET USING SE-RESNEXT WITH CASCADED AUTOENCODERS" for the Bulletin of Electrical Engineering and Informatics, 2025
45. Reviewer of "DYNAMIC MULTIOBJECTIVE OPTIMIZATION FOR COMPLEX INTERDEPENDENT AIRSPACE CONFLICT RESOLUTION" for the IEEE Transactions on Intelligent Transportation Systems, 2025
46. Reviewer of "ADAPTIVE REINFORCEMENT LEARNING FOR SECURE AND OPTIMIZED DECISION MAKING IN VEHICULAR NETWORKS" for the IEEE Transactions on Intelligent Transportation Systems, 2025

47. Reviewer of "ML-DRIVEN DEVOPS: AN EMPIRICAL FRAMEWORK FOR PREDICTIVE OPTIMIZATION AND INTELLIGENT AUTOMATION IN CI/CD PIPELINES" for the Software: Practice and Experience, 2025
48. Reviewer of "A SELF-LEARNING FRAMEWORK FOR ADAPTIVE FREE-FORM PIPE ROUTING IN AEROENGINES" for the Science China Technological Sciences, 2025
49. Reviewer of "ENERGY-EFFICIENT CLUSTERING AND ROUTING USING GREEN PHOTOSYNTHESIS AND RESPIRATORY-BASED OPTIMIZATION AND ADAPTIVE PARROT OPTIMIZATION FOR ENHANCED QOS IN MANETS" for the International Journal of Communication Systems, 2025
50. Reviewer of "FEDERATED DEEP LEARNING FOR SECURE 6G VIRTUAL NETWORK FUNCTIONS: ACHIEVING THE BALANCE BETWEEN DATA PRIVACY AND SECURITY MONITORING" for the International Journal of Communication Systems, 2025
51. Reviewer of "MANETS AND WSNS IN HYBRID GENGHIS KHAN SHARK AND LOTUS EFFECT OPTIMIZER FOR SECURE ROUTING" for the International Journal of Communication Systems, 2025
52. Reviewer of "ENHANCING THE SECURITY OF INTRUSION DETECTION SYSTEMS IN MANET USING FUZZY-BASED ARTIFICIAL IMMUNE SYSTEMS" for the International Journal of Communication Systems, 2025
53. Reviewer of "AN INTEGRATED ECCMGO AND MPCOA OPTIMIZATION FRAMEWORK WITH ASTRA-BASED INTRUSION DETECTION FOR SECURE MANET COMMUNICATION" for the International Journal of Communication Systems, 2025
54. Reviewer of "HYBRID ELK HERD GREEN ANACONDA-BASED MULTIPATH ROUTING AND DEEP LEARNING-BASED INTRUSION DETECTION IN MANET" for the International Journal of Communication Systems, 2025
55. Reviewer of "KOOKABURRA-OPTIMIZED SLIMMABLE NEURAL ROUTING FOR ADAPTIVE TRAFFIC CONTROL AND LIFETIME MAXIMIZATION IN MANETS" for the International Journal of Communication Systems, 2025
56. Reviewer of "A FEDERATED LEARNING-ENABLED SECURE AND SCALABLE SDN FRAMEWORK FOR ENERGY-EFFICIENT VANETS" for the International Journal of Communication Systems, 2025
57. Reviewer of "RAPID DETECTION OF COMMON SCAB, POWDERY SCAB, AND ENLARGE LENTICELS IN POTATO TUBERS USING DEEP LEARNING" for the Pest Management Science, 2025

58. Reviewer of "6G DEEP FEDERATED OPTIMIZATION FOR INTELLIGENT DATA ROUTING AND MONITORING SYSTEM USING GAME THEORY" for the International Journal of Communication Systems, 2025
59. Reviewer of "ENERGY-EFFICIENT AND SECURE ROUTING IN MANETS THROUGH EXPLICIT FEATURE INTERACTION GRAPH NEURAL NETWORKS" for the International Journal of Communication Systems, 2025
60. Reviewer of "MULTI-CRITERIA DECISION-MAKING (MCDM) FRAMEWORK: NON CONVOLUTIONAL RETURN-ALIGNED DECISION TRANSFORMER GRAPH NEURAL NETWORKS FOR ROUTING IN SECURE MANET-IOT 6G ENVIRONMENTS" for the International Journal of Communication Systems, 2025
61. Reviewer of "6G DEEP FEDERATED OPTIMIZATION FOR INTELLIGENT DATA QUANTUM RESISTANT ROUTING AND MONITORING SYSTEM USING GAME THEORY" for the International Journal of Communication Systems, 2025
62. Reviewer of "OPTIMIZED EXPLAINABLE DEEP LEARNING AND FEDERATED LEARNING-BASED QOS MULTICAST TREE ROUTING IN MANETS WITH CF-MMIMO" for the International Journal of Communication Systems, 2025
63. Reviewer of "A MULTI-OBJECTIVE HYBRID FRACTIONAL PUZZLE OPTIMIZATION (MOHFPO) FRAMEWORK FOR PRIORITY-AWARE SCHEDULING IN DEVICE-TO-DEVICE COMMUNICATION" for the International Journal of Communication Systems, 2025
64. Reviewer of "ASYNCHRONOUS FEDERATED REINFORCEMENT LEARNING FOR SCALABLE LOAD BALANCING IN SOFTWARE-DEFINED NETWORKS" for the International Journal of Communication Systems, 2025
65. Reviewer of "ENHANCED MANET SECURITY FRAMEWORK COMBINING CONVOLUTIONAL TRANSFORMER, QUANTUM SVM, AND ENERGY-AWARE CLUSTER HEAD SELECTION" for the International Journal of Communication Systems, 2025
66. Reviewer of "A Novel Air-to-Ground Communication Scheme for Advanced Big Data Collection in Smart Farming using UAVs" for the IEEE Access, 2024
67. Reviewer of "HyPE: Online Hybrid Pseudo-Bayesian Estimation Method for S-ALOHA-Based Tactical FANETs" for the IEEE Access, 2024
68. Reviewer of manuscript for the Wireless Personal Communications, 2024
69. Reviewer of manuscript ID ACCESS-2024-15337 for the IEEE Access, 2024
70. Reviewer of manuscript ID ACCESS-2024-46999 for the IEEE Access, 2024

71. Reviewer of "An Effective, Efficient and Scalable Link Discovery (EESLD) Framework for Hybrid Multi-controller SDN Networks" for the IEEE Access, 2023
72. Reviewer of "PRIO-AODV: A New Priority-Aware Routing Protocol for MANETs" for the IEEE Access, 2023
73. Reviewer with Engineering, Technology & Applied Science Research, 2023
74. Reviewer of "LINK-GUARD: an effective and scalable security framework for link discovery in SDN networks" for the IEEE Access, 2022
75. Reviewer of "An energy-aware and predictive fuzzy logic-based routing scheme in flying ad hoc networks (FANETs)" for the IEEE Access, 2021
76. Reviewer of "Topology Independent Multipath Routing for Data Center Networks" for the IEEE Access, 2021
77. Reviewer of "Centralized Intelligent Transport System using Multipath Routing Protocol for Vehicular Ad hoc Networks" for the IEEE Access, 2020
78. Reviewer of "Performance Analysis of Routing Protocols for UAV Communication Networks" for the IEEE Access, 2020
79. Reviewer of "A multi-path routing protocol based on link lifetime and energy consumption prediction for mobile edge computing" for the IEEE Access, 2020
80. Reviewer of manuscript ID ACCESS-2020-54638 for the IEEE Access, 2020
81. Reviewer of manuscript ID ACCESS-2020-22115 for the IEEE Access, 2020
82. Reviewer of manuscript ID ACCESS-2020-06767 for the IEEE Access, 2020
83. Reviewer for Information Security Journal: A Global Perspective, 2019
84. Reviewer of Journal of King Saud University - Computer and Information Sciences November 2018
85. Reviewer of The Seventh ICT International Student Project Conference (ICTISPC 2018) July 2018
86. Reviewer of 15th International Joint Conference on Computer Science and Software Engineering (JCSSE2018) July 2018
87. Reviewer of "EFAV-MERD: Expected Forwarding Area Volume and Residual Distance Mathematic Expectation Routing Protocol for UASNs" for the Ad Hoc Networks, 2016
88. Reviewer for Applied Computing and Informatics: "Fuzzy Bacterial Foraging Optimization Zone Based Routing (FBFOZBR) protocol for VANET", 2016

89. Reviewer of manuscript ROB-REG-15-0286 "An Improved FastSLAM 2.0 Algorithm Based on FC&ASD-PSO" for the Robotica, 2015
90. Manuscript Reviewer for International Journal of Digital Signal and Image Processing, 2014
91. Reviewer for International Conference on Computer & Information Sciences 2014 (ICCOINS2014), 2014

RESEARCH ACTIVITIES

Role	Number of Research University Grant	Number of Contract Research Grant	Total
Project Leader	6	3	9
Member	9	16	25

INTERNATIONAL MATCHING GRANT

1. Member, MG2 - 34.0 : Holistic and Smart Society Transformation : Artificial Intelligence in Utilization for Smart Village Development (Vot No. 1U001). Grant awarded: RM45,643.61 (August 2023 to July 2025).
2. Member, MG2 - 34.1 : Holistic and Smart Society Transformation: Enhancing Decision-Making Culture with Data-Driven Predictive Models in Smart Village Planning (Vot No. 04M58). Grant awarded: RM43,000 (August 2023 to July 2025).

GOVERNMENT AGENCY GRANT

1. Project Leader, Internet of Things for ecotourism and heritage properties monitoring in Petra Jaya, Kuching(Vot No. 1R052) Jabatan Tanah Dan Survei Sarawak. Grant awarded: RM38,000 (July 2024 to June 2026).
2. Member, Swk 13.0 : Space Syntax Approach In Achieving A Sustainable Quality Of Life In The Growth Of City In Relation To Urban Design, Housing And Architectural Development And Natural Aspects Of Heritage Based Tourism Attraction: The Case Of Petra Jaya, Kuching (Vot No. 1R061) Jabatan Tanah Dan Survei Sarawak. Grant awarded: RM250,000 (July 2024 to June 2026).
3. Member, SWK 13.1 : Study On Centrality Potential On Smart Infrastructure Via Space Syntax Network Analysis Of The Strategic Sustainable Growth Development For Petrajaya, Kuching, (Vot No. 1R061) Jabatan Tanah Dan Survei Sarawak. Grant awarded: RM170,000 (July 2024 to June 2026).

FLAGSHIP CENTRE OF EXCELLENCE/RESEARCH GROUP GRANT (COE/RG)

1. Project Leader, CoE/RG 1.4 Pembangunan Sistem Pemantauan Dan Peramalan Penyelenggaraan Seni Bina Warisan Berasaskan IoT(Vot No. 10G12) Flagship Centre of Excellence/ Research Group Grant. Grant awarded: RM48,000 (November 2022 to November 2025).
2. Member, CoE/RG 1.0 Kajian Seni Bina Dan Tarikan Pelancongan Berasaskan Warisan Bandar Di Bandaraya Johor Bahru. (Vot No. 10G07) Grant awarded RM 350,000 (November 2022 to November 2025).

CONTRACT RESEARCH GRANT (DTD)

1. Project Leader, Design And Development Of A Smart City Framework For Early Detection Of Water Intake And Quality (Vot No.PY/2020/05511) I Net Spatial Sdn. Bhd. Grant awarded: RM 77,700 (January 2021 to December 2021).
2. Project Member, Study of Sharia Compliant Digital Coin(Vot No.PY/2018/03492) OK Blockchain Centre Sdn. Bhd. Grant awarded: RM150,000 (October 2018 to September 2019).

COLLABORATIVE RESEARCH GRANT (CRG)

1. Project Leader, CRG 1.1 To Develop Blockchain Model In Managing Transportation And Logistic Resources (Vot No.PY/2018/02743) UTHM, UTeM dan HS Cargo Express (M) Sdn Bhd.Grant awarded: RM 40,000 (October 2018 to March 2021).

RESEARCH UNIVERSITY GRANT (RUG)

1. Project Leader, The Mobile Ubiquitous Positioning for Unmovable Physical Facility Tracking(Vot No. 02J56) Tier 2 RUG. Grant awarded: RM40,000 (April 2011 to March 2012).
2. Project Leader, Intelligent Transport System: The Prototype of Movable Robot Control Unit Based on Brainwave Signal Input(Vot No. 08J05) Tier 2 RUG. Grant awarded: RM32,000 (January 2013 to December 2013).
3. Project Leader, The Prototype of Smart Wheelchair Navigation and Controller Using non-Invasive Brain Signal Interface With FM Radio(Vot No. 05H79) Tier 1 RUG. Grant awarded: RM27,000.00 (April 2014 to March 2015).

4. Project Leader, The Prototype of Indoor Tracking Systems based on FM Radio(Vot No. 10J10) Tier 2 RUG. Grant awarded: RM24,000 (April 2014 to June 2015).
5. Project Leader, Biometric Tracking System For Immigrant Worker In Malaysia Using Ecg And Heart Sound (Vot No. 13H47) Tier 1 RUG. Grant awarded: RM50,000 (July 2016 to June 2018).
6. Project Leader, The Prototype of Wearable Wireless Sensing for Specific Human Motions Disorder (Vot No. PY/2017/01810) Tier 1 RUG. Grant awarded: RM50,000 (February 2018 to January 2020).
7. Member, Development Of Energy Efficiency System Using Wireless Sensor And Actuator Network(Vot No. 00H61) Tier 1 RUG. Grant awarded: RM147,000 (April 2011 to March 2013).
8. Member, Development Of Real-Time Malaysian Car Plate Number Recognition Algorithm(Vot No. 08J35) Tier 2 RUG. Grant awarded: RM32,000 (December 2012 to December 2013).
9. Member, Energy-Efficient and Precise Artificial Lighting System for Indoor Farming (Vot No. 03H05) Tier 1 RUG. Grant awarded: RM120,000 (May 2013, to April 2014).
10. Member, Improving The Reliability Of Digital Mobile Nodes Communications For Community-Centric Commuting Services(Vot No. 01G98) Flagship Grant. Grant awarded: RM 50,000 (April 2014 to March 2015).
11. Member, Computer-Assisted Emergency Response For Missing Person Incidents Within Indoor Environment Based On Strategic Informations Dissemination And Crowdsourcing(Vot No. 12J17) Tier 2 Grant awarded: RM 20,000 (October 2016 to December 2017).
12. Member, Internet Of Things Enabled Vertical Farming For Medicinal Plants And High Value Horticulture Commodity (Vot No. 14H74) Tier 1 Grant awarded: RM 50,000 (July 2016 to June 2018).
13. Member, The Technique Of Gait Recognition System Based On Anomaly Algorithm To Predict Freezing Of Gait (Fog) In Parkinson Disease (Vot No. 15H94) Tier 1 Grant awarded: RM 40,000 (October 2016 to September 2018).
14. Member, Prototype of Android 6.0 Indoor Tracking Systems based on Intelligent Fusion WiFi/IMU Signals (Vot No. 13H60) Tier 1 Grant awarded: RM 50,000 (July 2016 to September 2018).
15. Member, A Prototype Clinical Classification Of Heart Sound Signals Using Autoregressive Features And Hidden Markov Model (Vot No. 19H80) Tier 1 Grant awarded: RM 40,000 (February 2018 to December 2019).

**CONTRACT RESEARCH PROJECT AWARDED BY MINISTRY OF HIGHER EDUCATION
FUNDAMENTAL RESEARCH GRANT SCHEME (FRGS FUND)**

1. Project Leader, A Fundamental Study of Colour Code Scheduling in Beacon Selection Framework for Mobile Positioning in Multiblock Workspace(Vot No. PY/2014/04085). Grant awarded: RM77,100 (December 2014 to November 2016).
2. Project Leader, Basic Study of Optimization Techniques on sampling-based Motion Planning Algorithm for Closed Kinematic Chains Manipulator(Vot No. 3F530), Grant awarded: RM 44,000.00 (March 2010 to March 2012).
3. Member, Enhanced Scalable Routing Algorithm for Heterogeneous Underwater Linear Wireless Sensor Networks to Improve Pipelines Monitoring Coverage.(Vot No. PY/2015/05562), Grant awarded: RM75,200 (November 2015 to November 2017)
4. Member, An Enhanced Channel Assignment Scheme For Saturated Backhaul Wireless Mesh Networks To Mitigate Interference. (Vot No. PY/2015/05464), Grant awarded: RM73,200 (November 2015 to November 2017).
5. Member, Anomaly-based Algorithm for Gait Analysis to Predict Freezing of Gait (FOG) in Parkinson's Disease. (Vot No. PY/2016/07106), Grant awarded: RM63,500 (August 2016 to July 2018)
6. Member, Bidirectional traffic and intersection aware of TDMA MAC Protocol for VANET(Vot No. PY/2018/03844) Grant awarded: RM 73,000 (January 2019 to December 2020).

**CONTRACT RESEARCH PROJECT AWARDED BY MINISTRY OF HIGHER EDUCATION
-PROTOTYPE RESEARCH GRANT SCHEME (PRGS)**

1. Member Sustainability Prototype for Retrofitting Commercial Office Building Through Post Occupancy Evaluation (E-RETROW) (Vot No. 4L673) PRGS.Grant awarded: RM 120,000 (4 October 2016, to 31 July 2018)
2. Member, Pilot Deployment of Electronic Monitoring and Automation System System for Re-circulating Aquaculture System Using Wireless Embedded Technology (Vot No. 4L604) PRGS.Grant awarded: RM 300,000 (October 2011, to September 2013)

**CONTRACT RESEARCH PROJECT AWARDED BY MINISTRY OF SCIENCE &
TECHNOLOGY (SCIENCEFUND)**

1. Member, Mobile Ad Hoc Network Hop Routing Algorithm(Vot No. 3S258) ScienceFund.Grant awarded: RM 112,300 (December 2007 to June 2009).
2. Member, Friend-Assisted Secure Autonomous Mobile Ad Hoc Networks(Vot No. 3S257) ScienceFund.Grant awarded: RM 109,300 (December 2007 to June 2009).
3. Member, Node Protection Against Malicious Accusation Attack in Mobile Ad- Hoc Network (MANET)(Vot No. 79207) ScienceFund. Grant awarded: RM 168,000 (September 2007 to February 2009).
4. Member, 3D Scene Management to Reduce the Computational Cost of an Immerse Driving Simulator System(Vot No. 79282) ScienceFund. Grant awarded: RM 292,450 (April 2008 to September 2009).
5. Member, Development of Web-based Repository and Simulation System to Support Robotics E-Learning(Vot No. 79318) ScienceFund. Grant awarded: RM 179,300 (April 2008 to November 2010).
6. Member, Development of Wireless Hybrid Smart Sensor Network for Monitoring and Controlling Weather and Soil Factors on E-Agriculture (Vot No. 79305) ScienceFund.Grant awarded: RM 184,300 (April 2008 to November 2010).
7. Member, Database Summarization For Mobile Web Pre-Fetching (Mods) Using Rough Set (Vot No. 4S177) ScienceFund. Grant awarded RM143,000 (March 2015 to February 2017).

CONTRACT RESEARCH PROJECT AWARDED BY STATE GOVERNMENT

1. Member, Kreativiti Dan Inovasi Melalui Pengaturcaraan Liga Robotik Icare UtmKejora : Kemahiran Penyelesaian Masalah (Vot No. PY/2015/04738), Lembaga Kemajuan Johor Tenggara (KEJORA).Grant awarded: RM 80,000 (June 2015, to May 2017)
2. Member, New Era of Green Innovation & Sustainability 2016 (NEGISCO '16) Phase 1 - Wealth Creation Lab (Vot. No. PY/2016/07767), Networking Grant. Grant awarded RM 46,613.00 (September 2016 – October 2016)

CONSULTATION)

1. Project Leader, [Sistem Pengurusan Maklumat Bersepadu (Spmb) Akademik Pencegahan Rasuah Malaysia (MACA)] (Uni-Technologies Sdn Bhd (UTSB) Vol No. 1683) Client: Akademi Pencegahan Rasuah Malaysia (MACA), Suruhanjaya Pencegahan Rasuah Malaysia (SPRM). (08/05/2020 to 07/04/2024).

2. Project Member, Pembangunan Sistem Bantuan Kewangan Orang Kurang Upaya (BKOKU) (Uni-Technologies Sdn Bhd (UTSB) Vot No. 2246) Client: Kementerian Pendidikan Tinggi (KPT). (08/05/2023 to 07/11/2024).
3. Project Member, Kajian Kebolehlaksanaan Pembangunan Sistem Pengurusan Maklumat Bersepadu MRSM (MYMRSM) (Uni-Technologies Sdn Bhd (UTSB) Vot No. 2035) Client: Majlis Amanah Rakyat (MARA). (28/08/2023 to 29/01/2024).
4. Project Member, International Conference On Student And Disabled Student Development (Hal Ehwal Mahasiswa UTM (HEMA) Vot No. A.J060000.5600.07673) Client: Open Participant. (29/11/2019 to 01/12/2019).
5. Project Member, International Conference On Student Development (ICOSD) 2023 (Pejabat Naib Canselor Vot No. A.J0.6000.5600.07673) Client: Jabatan Timbalan Naib Canselor (Hal Ehwal Pelajar Dan Alumni) Universiti Teknologi Malaysia (UTM). (01/01/2022 to 29/02/2024).
6. Project Member, Training For Professional Certification In Blockchain Industry Proficiency (Uni-Technologies Sdn Bhd (UTSB) Vot No. 2332) Client: ICT@JOHOR . (06/09/2023 to 12/12/2023).
7. Project Member, Pakar Bidang @ Subject Matter Expert (SME) - Blockchain (Uni-Technologies Sdn Bhd (UTSB Vot No 2078.) Client: Timbalan Naib Canselor (Pembangunan). (01/03/2022 to 31/12/2025).
8. Project Member, Projek Rejuvenasi Desa Bakti Universiti Teknologi Malaysia (REDBAK) (Centre For Real Estate Studies (CRES) Vot No. A.J090903.5500.07197) Client: Timbalan Naib Canselor (Pembangunan) . (01/01/2022 to 30/09/2022).
9. Project Member, Pelaksanaan Penyelenggaraan E-IPTS (Uni-Technologies Sdn Bhd (UTSB) Vot No. 1964) Client: Kementerian Pengajian Tinggi Malaysia. (02/01/2021 to 01/01/2022).
10. Project Member, Penyelenggaraan Sistem E-IPTS (Uni-Technologies Sdn Bhd (UTSB) Vot No.2222) Client: Jabatan Pendidikan Tinggi, Kementerian Pendidikan Tinggi. (2/1/2023 to 1/1/2024).
11. Project Member, Pembaharuan Lesen Gmail Account Untuk Yayasan Pelajaran Johor Tahun 2022 (Uni-Technologies Sdn Bhd (UTSB) Vot No. 2137) Client: Yayasan Pelajaran Johor. (duration).
12. Project Member, PENYELENGGARAAN SISTEM E-IPTS (Uni-Technologies Sdn Bhd (UTSB) Vot No. 2043) Client: Bahagian Standard Swasta, Jabatan

Pendidikan Tinggi, Kementerian Pengajian Tinggi Malaysia. (02/01/2022 to 01/01/2023).

KEYNOTE AND INVITED SPEAKER

1. Keynote Speaker at Hybrid Seminar “Developing Digital Product Innovation for Indonesian SME Industry in 2025”, 23rd January 2025.
2. Keynote Speaker at the 9th *International Conference on Informatics and Computing* (ICIC) 2024 Hotel Santika Premiere Dyandra Hotel & Convention Medan Jl. Kapten Maulana Lubis No. 7 Petisah Tengah Kec. Medan Petisah Kota Medan 20111, Sumatera Utara-Indonesia
3. Invited Speaker at the 3rd International Conference HZU and UTM 2018, Hui Zhou University, Hui Zhou China.
4. Invited Speaker at the International Seminar Between HZU and UTM 2016, Hui Zhou University, Hui Zhou China.

CONFERENCE PRESENTER AND CHAIRPERSON

1. Presenter at the 4th International Conference on Smart Sensors and Application (ICSSA 2022)(International)
2. Presenter at The 6th International Conference of Reliable Information and Communication Technology 2021 (IRICT 2021) (International)
3. Presenter at the 4th International Conference on Science, Engineering and Social Sciences (ICSESS 2021) (International)
4. Presenter at the International Conference of Student and Disabled Student Development 2019 (ICOSD 2019) (International)
5. Presenter at the 11th Asian Conference on Intelligent Information and Database Systems (ACIIDS 2019) (International)
6. Presenter on "Smart Campus Initiative" at the University ICT Administration Conference 2019 (PPICTP2019) (National)
7. Chairperson for a Parallel Session at the International Student and Disabled Student Development 2019 (International)
8. Chairperson at the National Student Development Conference (NASDEC 2018) (National)

9. Presenter at the 3rd International Joint Seminar between Huizhou University and Universiti Teknologi Malaysia (International)
10. Presenter at the National Student Development Conference (NASDEC) 2016 (National)
11. Presenter at the 1st International Joint Seminar on Multidisciplinary Method and Model(International)
12. Presenter at The International Conference on Science in Information Technology (ICSITECH)(International)
13. Presenter at the 1st International Conference on Science, Technology, and Interdisciplinary Research (IC-STAR) (International)
14. Presenter at The 7th Asian Conference on Intelligent Information and Database Systems (ACIIDS 2015) (International)
15. Presenter at the International Student Management Seminar (University)
16. Chairperson at the Postgraduate Annual Research Seminar (PARS 2010) (Faculty)
17. Chairperson at the National Student Development Conference 2010 (NASDEC 2010) (National)
18. Presenter at the IUSRCE University Social Responsibility Conference 2010 (International)
19. Chairperson at the IUSRCE International University Social Responsibility Conference and Exhibition 2010 (International)

TEACHING ACTIVITIES

Semester	Sem	Subject Code	Subject	Credit Hour	Total Student
2024/2025	2	MECS2323	Advanced Computer Network & Cloud Computing	3	7
2024/2025	1	MECS2323	Advanced Computer Network & Cloud Computing	3	14
2023/2024	2	MECS2323	Advanced Computer Network & Cloud Computing	3	7
2022/2024	1	SECR1213	Network Communication	3	42
2022/2023	1	SECR1213	Network Communication	3	37

2021/2022	1	SECR1213	Network Communication	3	43
2020/2021	2	MCSS2313	Adv. Comp. Sys. & Architecture	3	11
2020/2021	2	SECR2033	Computer Org. & Architecture	3	38
2020/2021	1	MCSS2313	Adv. Comp. Sys. & Architecture	3	6
2019/2020	2	MCSS2313	Adv. Comp. Sys. & Architecture	3	4
2019/2020	1	SCSR2033	Computer Org. & Architecture	3	20
2018/2019	1	MCSS2313	Adv. Comp. Sys. & Architecture	3	6
2017/2018	2	MCSS2313	Adv. Comp. Sys. & Architecture	3	6
2017/2018	1	MCSS2313	Adv. Comp. Sys. & Architecture	3	8
2016/2017	2	MCSS2313	Adv. Comp. Sys. & Architecture	3	3
2016/2017	1	MCSS2313	Adv. Comp. Sys. & Architecture	3	12
2015/2016	1	SCSR1013	Logic Digital	3	14
2014/2015	2	SCSR1213	Network Communication	3	22
2014/2015	1	SCSR3104	Application Development	4	30
2014/2015	1	SCSR3104	Application Development	4	12
2014/2015	1	SCR3104	Application Development	4	1
2013/2014	2	SCSJ1013	Programming Technique I	3	1
2013/2014	2	SCSR1213	Network Communication	3	18
2013/2014	1	MCC2313	Adv. Comp. Sys. & Architecture	3	8
2012/2013	2	MCC2313	Adv. Comp. Sys. & Architecture	3	20
2012/2013	1	MCC2313	Adv. Comp. Sys. & Architecture	3	34
2012/2013	1	MCS2070	Information Security Seminar	0	9
2011/2012	2	MCC2313	Adv. Comp. Sys. & Architecture	3	19
2011/2012	1	SCR2033	Computer Org. & Architecture	3	30
2011/2012	1	SCR1043	Computer Org. & Architecture	3	9
2011/2012	1	SCE3103	Computer Org. & Architecture	3	12
2011/2012	1	SCJ2033	Computer Org. & Architecture	3	29
2010/2011	2	SCJ2103	Application Development	3	40
2010/2011	1	SCR1043	Computer Org. & Architecture	3	52
2009/2010	2	SCJ2103	Application Development	3	29
2009/2010	1	SCJ1013	Programming Technique I	3	35
2008/2009	2	SCJ2103	Application Development	3	32
2008/2009	1	SCK1013	Comp. Org. & Assembly Lang.	3	9
2008/2009	1	SCR1043	Computer Org. & Architecture	3	28
2007/2008	3	SCR2043	Operating System	3	27
2007/2008	2	SCJ1023	Programming Technique II	3	18
2007/2008	1	SCE3103	Computer Org. & Architecture	3	32
2007/2008	1	SCJ1013	Programming Technique I	3	20
2006/2007	2	SCK1023	Digital Logic	3	2

SUPERVISION

PhD Student

Number of students graduated	Main supervision	Co-supervision
7	5	2

Year	No.	Name	Status	Title	Roles of Supervision
2011	1	Mohammad Taghi Kheirabadi	Graduated 8 April 2015	An Energy Efficient and Void Avoidance Algorithm for Underwater Sensor Network PressureBased Opportunistic Routing	Main Supervisor
2012	2	Mudassar Ahmad	Graduated 3 August 2016	TCP Congestion Control Mechanisms in Short and Long Distance Network	Co-Supervisor
2012	3	Amani Mahagoub Omer Elhag Elawad	Deceased 2012 – 2017	Database Security Privacy Preserving In Published Database	Main Supervisor
2013	4	Abdulaziz Yahya Yahya Al-Nahari	Graduated April 2019	Receiver-based AODV Routing Protocol for MANETs	Main Supervisor
2013	5	Wan Mohd Yaakob Wan Bejuri	Graduated November 2019	Indoor Positioning Algorithm Using Gyroscpe Sensors for Parkinson	Main Supervisor

				Patient	
2015	6	Muhamad Tayyab	Graduated October 2020	Enhanced Reliable And Energy Efficient Pressure Based Data Forwarding Schemes For Underwater Wireless Sensor Networks	Co-Supervisor
2016	7	Nasarudin Bin Ismail	Graduated March 2021	Energy Efficient And Reliable Location Free Using Opportunistic Routing For Underwater Sensor Networks	Main Supervisor
2014	8	Siti Hadrina Sheikh Hussain	Graduated 2022	Classification Algorithm of Heart Sound Signal Processing	Main Supervisor
2016	9	Mohammed Abduljalil Mohammed AlSaeedi	Graduated 2025	A Scalable Software Defined Networking Controller for Internet of Things	Main Supervisor
2019	10	Siman Emmanuel	Graduated 2023	Enhanced Cluster-Based Multiple Access Adaptive Protocol For Vehicular Ad Hoc Network Using Modified Stable Clustering Algorithm	Co-Supervisor
2020	11	Chinoso Okereke	Graduated 2025	Autonomous Underwater Vehicle (AUV)	Main Supervisor

				For Monitoring And Tracking With Multiple Path Planning Via Machine Learning In Internet Of Underwater Things (IoUT)	
2020	12	Cindy Ujan	Ongoing 2025	A Modified Energy Efficient Swarm Intelligence-Inspired Algorithm for Software-Defined Wireless Sensor Networks	Main Supervisor
2020	13	Asma Basharat	Ongoing 2025	Framework for Secure Routing in Software Defined Networking	Main Supervisor
2020	14	Ruan Xiaole	Terimanted 2023		Main Supervisor
2021	15	Saad Mohsen Hassan	Ongoing 2025		Main Supervisor

MSc. Student

Number of students graduated	Main supervision	Co-supervision
5	3	2

Year	No.	Name	Status	Title	Type	Roles of Supervision
2010	1	Md Fadil bin Md Esa	Graduated May 2014	Bidirectional Swarm Based Path Planning for Mobile Robot	MSc. Research	Co-Supervisor
2010	2	Wan Mohd Yaakob Wan Bejuri	Graduated 10 September 2013	Colour Constancy Feature Detection and Matching for Wireless LAN/	MSc. Research	Main Supervisor

				Camera Indoor Positioning System		
2010	3	Mojtaba Alizadeh	Graduated 18 September 2012		MSc. Mix Mode	Co-Supervisor
2016	4	Raziman Bin Zakaria	Graduated 14 February 2018		MSc. Mix Mode	Main Supervisor
2019	5	BASHIR ABDULLAHI	Graduated November 2019		MSc. Mix Mode	Main Supervisor
2021	6	Ali Mohammed Ali	Graduated 2023		MSc. Mix Mode	Main Supervisor
2023	7	Nur Shahirah Binti Jailani	Graduated 2025		MSc. By Research	Main Supervisor

POSTGRADUATE EXAMINATION /VIVA

UTM STUDENTS VIVA - VIVA CHAIRPERSON

1. Nurul Nadzirah binti Mohd Hasri, Improved Funnel-Gene Set Enrichment Analysis Using Adaptive Elastic-Net Penalization Method To Identify Significant Gene Sets, May 2021
2. Devi Willieam Anggara, Enhancement Of Feature Extraction Method In Improving Marker Detection For Augmented Reality Environment, February 2021.

UTM STUDENTS VIVA - ASSISTANT VIVA CHAIRPERSON

1. Abdallah Elhigazi (Doctor of Philosophy) Wireless Local Area Network Management Frame Denial-of-Service Attack Detection and Mitigation Schemes, April 2020.
2. Ali Syari'ati Bin Mohd Salleh (Doctor of Philosophy) Hydrofloroether Impurities Chemical Detection Using Deep Learning Evolving Spiking Neural Network, October 2019.
3. Myasar Mundher Adnan (Doctor of Philosophy) Deep Learning Based for Automatic Image Annotation, October 2019.

4. Lim Kah Seng (Master by Research) Developing A Knowledge-Based Graphical Password Anti-Shoulder Surfing Mechanism For The Mobile Device, March 2012.
5. Hasan Chizari (Doctor of Philosophy) A Duality Approach For Energy Efficient Wireless Sensor Network Design, December 2011.
6. Ikhwan Zaini B. Awang Mat Saaid (Master by Research) Komponen Keselamatan Maklumat Institusi Pengajian Awam Malaysia, November 2010.
7. Roime Puniran (Master by Research) An Integration of Ant Algorithm and Travelling Salesman Problem for Peer-to-peer Routing Protocol in An Enterprise Network, April 2010.
8. Md Sah bin Hj Salam (Doctor of Philosophy) A Fusion of Statistical and Connectionist Approach for Segmentation of Connected Speech, September 2009.

POSTGRADUATE FIRST ASSESSMENT - EXAMINER

1. Bilal Ashfaq Ahmed (Doctor of Philosophy) Resource Allocation and Scheduling Framework Using Software Defined Networking for Edge as a Service In Internet of Things, October 2019.
2. Mazhar Javed Awan (Doctor of Philosophy) Fully Automated Early Knee Osteoarthritis Intelligent Decision Making of Magnetic Resonance Imaging, October 2019.
3. Muhammad Kaleem (Doctor of Philosophy) An Adaptive Thermal-Aware Routing Algorithm With Path Selection and Interval Record Keeping Mechanism October 2019.
4. Mustafa Sabah Taha El Tameemi (Doctor of Philosophy) An Improved Steganography Technique Using Contrast Level and Secret Message Encryption, November 2018.
5. Gul Sahar (Doctor of Philosophy) Energy Efficient Data Acquisition Technique in Large-Scale Wireless Sensor Networks, November 2018.
6. Abdullah Bin Mat Safri (Doctor of Philosophy) Enhanced Predictive PreHandover Scheme For Urban Area Vehicular Networks, November 2018.
7. Nosiba Hisham Osman Khougali (Doctor of Philosophy) Design of a High Efficient Small Size Mobile Robot Based on Wireless Power Transmission, April 2018.
8. Abdulrahman Mohammed Alamin Abdulgasem (Doctor of Philosophy) Multi Modal Features for Content Based Image Retrieval, April 2014.

9. Muhammad Zahid Abbas (Doctor of Philosophy) A Novel Congestion Prediction, Avoidance and Management Technique for Clustered Base WMN, April 2014.
10. Khalid Haseeb (Doctor of Philosophy) Energy Aware Multi-Layer Cluster Based Mechanism for Wireless Sensor Network, December 2013.
11. Reza Aghababseyan (Master by Research) Network Security of Smart Home System, November 2012.
12. Muhamad Gary Shafer (Master by Research) Monitoring and Control of Electric Power Usage Using Wireless Embedded Network, May 2011.
13. Yoanda Alim Syahbana (Master by Research) Quality Enhancement for Video Stream Over Wireless and Mobile Network, May 2011.

MSc PROJECT - EXAMINER

1. Malek Najib Abdul Kareem Omar, Authentication of Alert Dissemination on Friendship based Mechanism, December 2013.
2. Jatau Isaac Katuka, Mosaicing of Torn Image using Hybrid Method for Forensic Investigation, December 2013.
3. Ibrahim Thorig, High Inter-Vehicular Mobility Protocol to Handle Intersection Problem in Vehicular Ad Hoc Networks, December 2013.
4. Muhammad Syafiq Samsi, Identity Disintegration Mechanism for Cardiothoracic Information Privacy, December 2013.

MSc INTERNAL - EXAMINER

1. Zulwaqar Zain Bin Mohtar, Smart Contract Access Control Method In Blockchain-Based Distributed File System, October 2024.
2. Mossolihen bin Dasuki, Enhanced Load Balancing Algorithm for Internet of Things Blockchain Nodes, November 2021.
3. Eko Arip Winanto, Extension Proof Of Attack Signature In Collaborative Intrusion Detection System Using Blockchain Consensus Algorithm, May 2021.
4. Yoanda Alim Syahbana, Frame Alignment and Frame Insertion Algorithm to Measure Video Quality in Video Stream over Mobile Network, December 2011.

5. Muhamad Gary Shafer, Dual Demand Side Management (2DSM) Method in Optimizing Utilization of Available Power Supply, November 2013.

PhD INTERNAL EXAMINER

1. Zabid Bashar Ahmed Khalaf, Deep Reinforcement Learning Multi-Agent System Model For Link Flood Attack Detection and Mitigation Smart City Networks 2025.
2. Ajibade Lukuman Saheed, Mapreduce-Based Straggler Mitigation Scheme For Big Data Environment 2024.
3. Mohamed Mostafa Ali Montaser, A Data Explosion Reduction Algorithm For Blockchain-Based Tracking And Tracing Systems In Supply Chain Management 2024.
4. Siti Nurmaini, Embedded Interval Type-2 Neuro-Fuzzy Controller In Mobile Robot Navigation With Weightless Neural Network 2011.

PUBLICATIONS

Web of Science Core Collection

Total	Article and Review Article	Conference/Proceeding Paper	Correction	H-Index
35	17	16	2	9

Scopus-Indexed Analyze Author Output

Total	Articles Journal	Review Article	Conference/Proceeding Paper	Book Chapter	H-Index
79	35	4	30	3	13

CURRENT PUBLIC RESEARCH PROFILE (verified 24 August 2026)

ORCID: 0000-0002-1478-0138

Scopus Author ID: 16033644000 — 79 documents, 871 citations, h-index 13.

Web of Science ResearcherID: C-9887-2011 — 35 Core Collection publications, 359 citations, h-index 9.

Web of Science verified peer reviews: 183 (99th percentile at the date verified).

SELECTED CONSULTANCY AND INDUSTRY ENGAGEMENT

Digital government and enterprise systems — integrated higher-education data warehouse feasibility study (RM498,937.07); MyMRSM integrated information management system feasibility study (RM198,552.84); BKOKU financial-assistance system study (RM400,000); E-IPTS maintenance programme (RM749,840 combined recorded value); and Google Workspace licence renewal (RM69,800).

Blockchain capability and professional training — training and SME services for industry (RM108,000); professional certification training (RM199,672.20); and blockchain subject-matter expert engagement (RM100,000).

Web of Science-Indexed Articles and Review Articles [17] :

1. Junaidi A., Othman M.S.B., Hashim S.Z.M., Mohamad M.M., Kesa D.D., Nurfikri A., *Smart Villages: A Systematic Review of Trends, Models, and Metrics*, *Cogent Social Sciences*, Vol. 11, No. 1, 2025, Article 2492833, DOI: 10.1080/23311886.2025.2492833.
2. Junaidi A., Hashim S.Z.M., Bin Othman M.S., Bin Mohamad M.M., Alhussian H., Abdulkadir S.J., Nasser M., Bena Y.A., *Deep Learning and Edge Computing in Agriculture: A Comprehensive Review of Recent Trends and Innovations*, *IEEE Access*, Vol. 13, 2025, pp. 137464–137490, DOI: 10.1109/ACCESS.2025.3595621.
3. Hassan S.M., Mohamad M.M., Muchtar F.B., *Enhancing MANET Security Through Long Short-Term Memory-Based Trust Prediction in Location-Aided Routing Protocols*, *IEEE Access*, Vol. 13, 2025, pp. 120142–120168, DOI: 10.1109/ACCESS.2025.3572619.
4. Hassan S.M., Mohamad M.M., Muchtar F.B., Dawoodi F.B.P., *Enhancing MANET Security Through Federated Learning and Multiobjective Optimization: A Trust-Aware Routing Framework*, *IEEE Access*, Vol. 12, 2024, pp. 181149–181178, DOI: 10.1109/ACCESS.2024.3505236.
5. Hassan S.M., Mohamad M.M., Muchtar F.B., *Advanced Intrusion Detection in MANETs: A Survey of Machine Learning and Optimization Techniques for Mitigating Black/Gray Hole Attacks*, *IEEE Access*, Vol. 12, 2024, pp. 150046–150090, DOI: 10.1109/ACCESS.2024.3457682.
6. Alsaeedi M., Mohamad M.M., Al-Roubaiey A., *SSDWSN: A Scalable Software-Defined Wireless Sensor Networks*, *IEEE Access*, Vol. 12, 2024, pp. 21787–21806, DOI: 10.1109/ACCESS.2024.3362353.
7. Okereke C.E., Mohamad M.M., Wahab N.H.A., Elijah O., Al-Nahari A., Zaleha S.H., *An Overview of Machine Learning Techniques in Local Path Planning for Autonomous Underwater Vehicles*, *IEEE Access*, Vol. 11, 2023, pp. 24894–24907, DOI: 10.1109/ACCESS.2023.3249966.

8. Emmanuel S., Bin Isnin I.F., Bin Mohamad M.M., *A Reliable Merging Link Scheme Using Weighted Markov Chain Model in Vehicular Ad Hoc Networks*, *Sensors*, Vol. 22, No. 13, 2022, Article 4861, DOI: 10.3390/s22134861.
9. Alsaeedi M., Mohamad M.M., Al-Roubaiey A.A., *Toward Adaptive and Scalable OpenFlow-SDN Flow Control: A Survey*, *IEEE Access*, Vol. 7, 2019, pp. 107346–107379, DOI: 10.1109/ACCESS.2019.2932422.
10. Ismail N., Mohamad M.M., *Review on Energy Efficient Opportunistic Routing Protocol for Underwater Wireless Sensor Networks*, *KSII Transactions on Internet and Information Systems*, Vol. 12, No. 7, 2018, pp. 3064–3094, DOI: 10.3837/tiis.2018.07.006.
11. Al-Nahari A., Mohamad M.M., *Receiver-Based Ad Hoc On-Demand Multipath Routing Protocol for Mobile Ad Hoc Networks*, *PLOS ONE*, Vol. 11, No. 6, 2016, Article e0156670, DOI: 10.1371/journal.pone.0156670.
12. Al-Nahari A., Mohamad M.M., Zahilah R., *An Improved Maintenance Strategy in Ad Hoc On-Demand Routing Protocols for Mobile Ad Hoc Networks*, *Jurnal Teknologi*, Vol. 78, No. 6–11, 2016, pp. 59–65.
13. Bejuri W.M.Y.W., Mohamad M.M., Raja R.Z., Radzi M., *Optimization of Rao-Blackwellized Particle Filter in Activity Pedestrian Simultaneous Localization and Mapping (SLAM): An Initial Proposal*, *International Journal of Security and Its Applications*, Vol. 9, No. 11, 2015, pp. 377–390, DOI: 10.14257/ijisia.2015.9.11.35.
14. Bejuri W.M.Y.W., Mohamad M.M., *Performance Analysis of Grey-World-Based Feature Detection and Matching for Mobile Positioning Systems*, *Sensing and Imaging*, Vol. 15, No. 1, 2014, Article 95, DOI: 10.1007/s11220-014-0095-7.
15. Bejuri W.M.Y.W., Mohamad M.M., Sapri M., Rahim M.S.M., Chaudry J.A., *Performance Evaluation of Spatial Correlation-Based Feature Detection and Matching for Automated Wheelchair Navigation System*, *International Journal of Intelligent Transportation Systems Research*, Vol. 12, No. 1, 2014, pp. 9–19, DOI: 10.1007/s13177-013-0064-x.
16. Kheirabadi M.T., Mohamad M.M., *Greedy Routing in Underwater Acoustic Sensor Networks: A Survey*, *International Journal of Distributed Sensor Networks*, Vol. 2013, Article 701834, DOI: 10.1155/2013/701834.
17. Ghafoor K.Z., Abu Bakar K., Salleh S., Lee K.C., Mohamad M.M., Kamat M., Arshad M.M., *Fuzzy Logic-Assisted Geographical Routing Over Vehicular Ad Hoc Networks*, *International Journal of Innovative Computing, Information and Control*, Vol. 8, No. 7B, 2012, pp. 5095–5120.

Web of Science-Indexed -corrections [2] :

1. Hassan S.M., Mohamad M.M., Muchtar F.B., *Advanced Intrusion Detection in MANETs: A Survey of Machine Learning and Optimization Techniques for Mitigating Black/Gray*

Hole Attacks (Vol. 12, pp. 150046, 2024), IEEE Access, Vol. 13, 2025, p. 137850, DOI: 10.1109/ACCESS.2025.3595635.

2. Alsaeedi M., Mohamad M.M., Al-Roubaiey A., *SSDWSN: A Scalable Software-Defined Wireless Sensor Networks (Vol. 12, pp. 21787, 2024), IEEE Access, Vol. 13, 2025, p. 54284, DOI: 10.1109/ACCESS.2025.3554084.*

SCOPUS-Indexed Articles [35] :

1. Emmanuel S., Isnin I.F.B., Mohamad M.M.B., *A survey of TDMA-based MAC protocols for vehicular ad hoc networks, International Journal of Engineering and Advanced Technology, Vol. 8, No. 5, 2019, pp. 1247–1259, DOI: 10.35940/ijeat.E1178.0585C19.*
2. Wahyono I.D., Saryono D., Putranto H., Asfani K., Rosyid H.A., Sunarti, Mohamad M.M., Said M.N.H.B.M., Horng G.J., Shih J.-S., *Emotion Detection based on Column Comments in Material of Online Learning using Artificial Intelligence, International Journal of Interactive Mobile Technologies, Vol. 16, No. 3, 2022, pp. 82–91, DOI: 10.3991/IJIM.V16I03.28963.*
3. Alsaeedi M., Mohamad M.M., Al-Roubaiey A., *SSDWSN: A Scalable Software-Defined Wireless Sensor Networks, IEEE Access, Vol. 12, 2024, pp. 21787–21806, DOI: 10.1109/ACCESS.2024.3362353.*
4. Hassan S.M., Mohamad M.M., Muchtar F.B., Dawoodi F.B.Y.P., *Enhancing MANET Security Through Federated Learning and Multiobjective Optimization: A Trust-aware Routing Framework, IEEE Access, 2024, DOI: 10.1109/ACCESS.2024.3505236.*
5. Bejuri W.M.Y.W., Mohamad M.M., Sapri M., Rosly M.A., *Investigation of color constancy for ubiquitous wireless LAN/camera positioning: An initial outcome, International Journal of Advancements in Computing Technology, Vol. 4, No. 7, 2012, pp. 269–280, DOI: 10.4156/ijact.vol4.issue7.30.*
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